





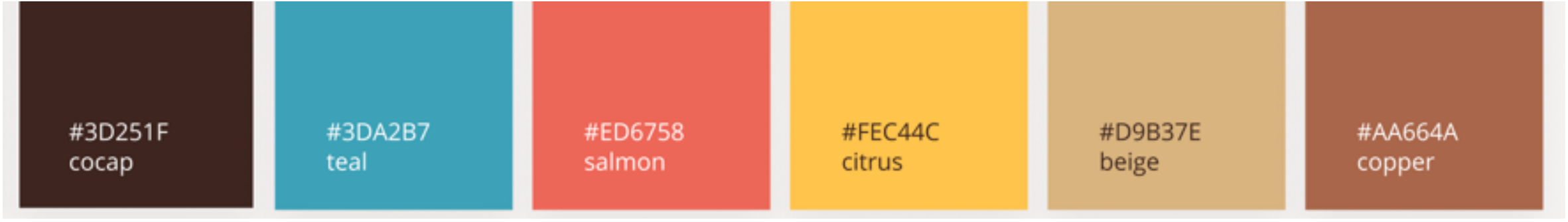
- **Prove that optimal transport exists, is computable, and behaves nicely**

**Handle both discrete and continuous data**

**Integrate costs over couplings**



measure theory tells us the "rules"



#3D251F  
cocap

#3DA2B7  
teal

#ED6758  
salmon

#FEC44C  
citrus

#D9B37E  
beige

#AA664A  
copper

# Coming “back” to OT

- We’re not doing this for the sake of math — we’re laying the foundation to:
  - Define what it means to compare two distributions
  - Move mass around in a well-defined, flexible way
  - **Prove that optimal transport exists, is computable, and behaves nicely**
- Measures let us:
  - **Handle both discrete and continuous data** in one framework
  - Split, push, and transform mass without ambiguity
  - **Integrate costs over couplings** — the heart of OT
- OT is *just* optimization over measures, **measure theory tells us the “rules”** to do so



# Gaspar Monge (1746-1818)



- Father of descriptive and differential geometry

666. MÉMOIRES DE L'ACADÉMIE ROYALE

*M É M O I R E*  
*S U R L A*  
*T H É O R I E D E S D É B L A I S*  
*E T D E S R E M B L A I S.*

Par M. M O N G E.

**L**ORSQU'ON doit transporter des terres d'un lieu dans un autre, on a coutume de donner le nom de *Déblai* au volume des terres que l'on doit transporter, & le nom de

